KUKA contact-energy guard: what the traces establish

Results-first adaptation of the original 2-DOF PoC Figures 3–8. The current KUKA reserve/governor policy completes the main blocking fixture and preserves contact, while the frozen endpoint filter exhibits a negative physics energy prefix and substantial contact disturbance. This is a report about the existing implementation; no controller, gains, energy rules or plant parameters were redesigned.

Run	End [s]	Net out [J]	Min raw E [J]	Force peak error [N]	Outcome
Main UF	18	0.241623	-0.1216233	0.397	complete
Main A	18	0.119999	-1.11e-06	53.104	complete
Main D	18	0.119591	0.0004092	1.241	complete
Historical D	18	0.119869	8.77e-05	1.045	complete
Tiny budget D	9.533	6.86e-05	0.0009314	0.066	stopped

Energy minima include every executed 0.25 ms physics increment. Force errors use the original 10 N tool/table task target and the declared task window, beginning at 8.71 s. A stopped run contributes only its executed samples; it has no inferred release or recovery.

KUKA label	Actual role
UF	Matched unfiltered reference; work accounts monitored
A	Frozen Stage 1 endpoint joint-torque energy QP
B	Physics-prefix energy checks and nonlinear replay
C	B plus finite backup reserve, virtual progression and backup selection
D	C plus explicit per-stage task acceptance and QP task rows

H and R receive the same human work. Archived C and D coincide, and D performs zero recorded QP solves across the 13-case matrix. The supported improvement is therefore evidence for the coupled reserve/governor/backup policy in these fixtures, without establishing active task-QP superiority, independent mismatch accounting or global robot passivity.

13-case matrix	Completed	Stopped	Energy-invalid	Recorded QP calls
UF	13	0	8	0
A	9	4	3	1354
B	5	8	0	10
C	12	1	0	0
D	12	1	0	0

Original scientific questions and KUKA coverage

Family	Status	Supported result and missing comparison
F3	Adapted	No-human task-work exemption/transparency; no closed-loop whole-port baseline
F4	Unsupported	No independently configured mismatch residual; show existing-account identity only
F5	Adapted	Executed blocking power, net/gross work and raw energy for actual KUKA policies
F6	Unsupported	No implemented p_delta separation; combine the supported identity diagnostic with F4
F7	Adapted	One continuous help-then-block history; uncapped charging
F8	Adapted	Full coupled-policy interaction/recovery; joint-torque intervention replaces scalar command

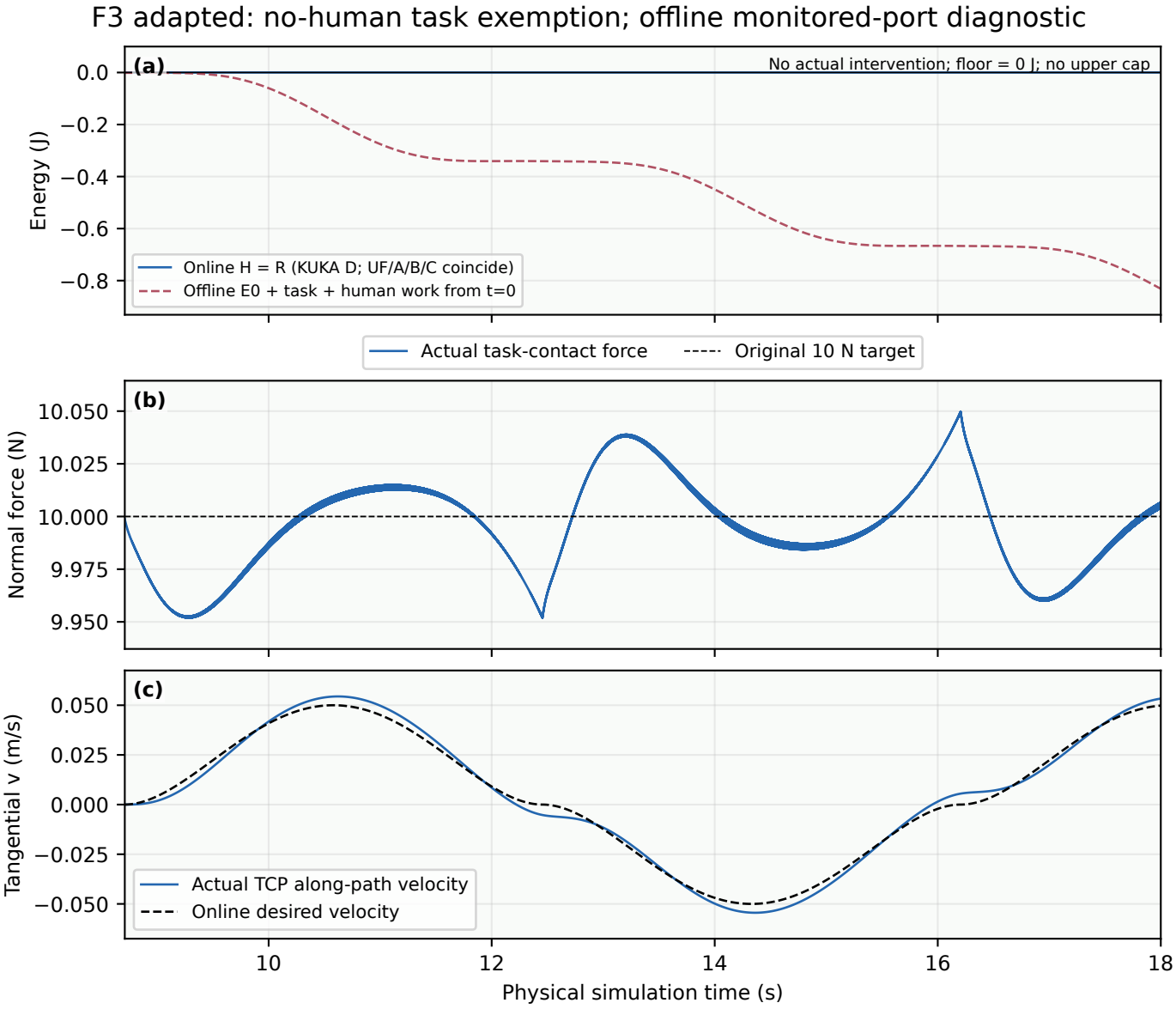
No family is an exact controller/scenario reproduction. The original 2-DOF plots and raw data establish each question and panel meaning; every result curve here is KUKA data. The complete file/function, signal, units, configuration, adaptation and source-line mapping is recorded in figure_crosswalk.csv.

2-DOF paper label	Actual implementation and relation to KUKA
C0	Nominal; closest experimental role is KUKA UF
C1	Whole-port QP; no KUKA closed-loop equivalent
C2	Residual-only QP; no independent KUKA mismatch/residual equivalent
C3	C3_dual_ledger_qp with governor OFF
C4	The same C3_dual_ledger_qp with governor ON

The historical code mode C4_dual_ledger_safe_scalar is a separate diagnostic and is not paper C4. KUKA C/D labels are unrelated to that naming system. KUKA B versus D changes reserve/governor, backup selection and task protection; it cannot isolate a governor effect.

The main human force acts at a body-fixed link5 point; elbow fixtures use link3. Human power uses velocity at that actual moving point. Tool/table normal contact, its 10 N target and TCP tracking are separate task signals. Positive human power enters the robot; negative power is robot-to-human output. Known force, simulated contact and finite sampled tests remain explicit assumptions.

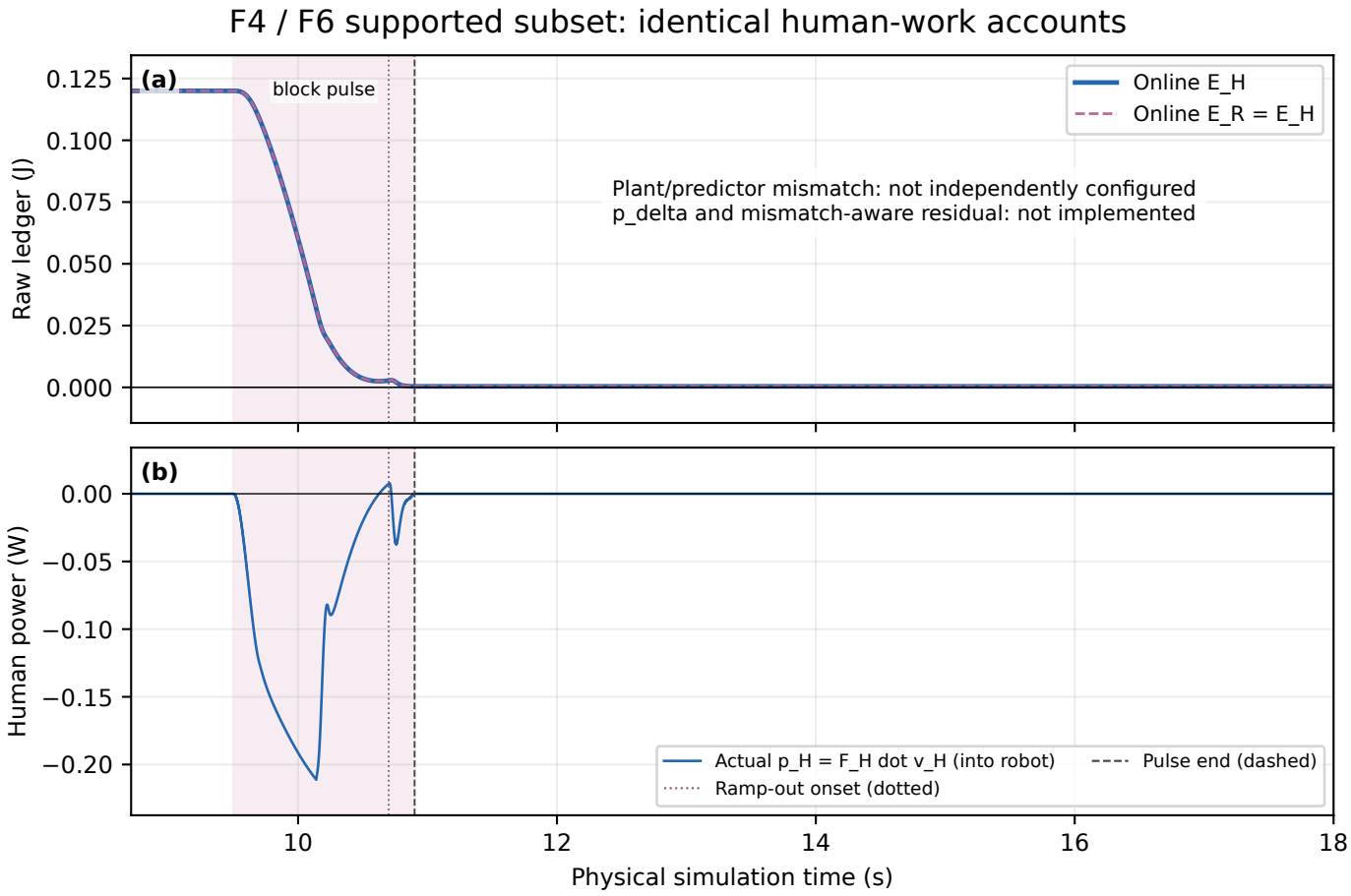
F3 — Task work is exempted, with the causal gap visible



F3. No-human task behavior and accounting. The matched no-human traces show actual ledgers, tool/table normal force and along-path velocity with their implemented floor and references. The offline monitored-port integral starts at t = 0 and includes startup task/contact work before the displayed sliding window, so its initial offset is not attributed solely to useful sliding.

Zero-initial-energy D preserves the task with E_H = E_R = 0 and no actual intervention. The offline diagnostic is not a command-generating whole-port controller, so the original causal claim that useful friction would stop such a controller remains untested for KUKA.

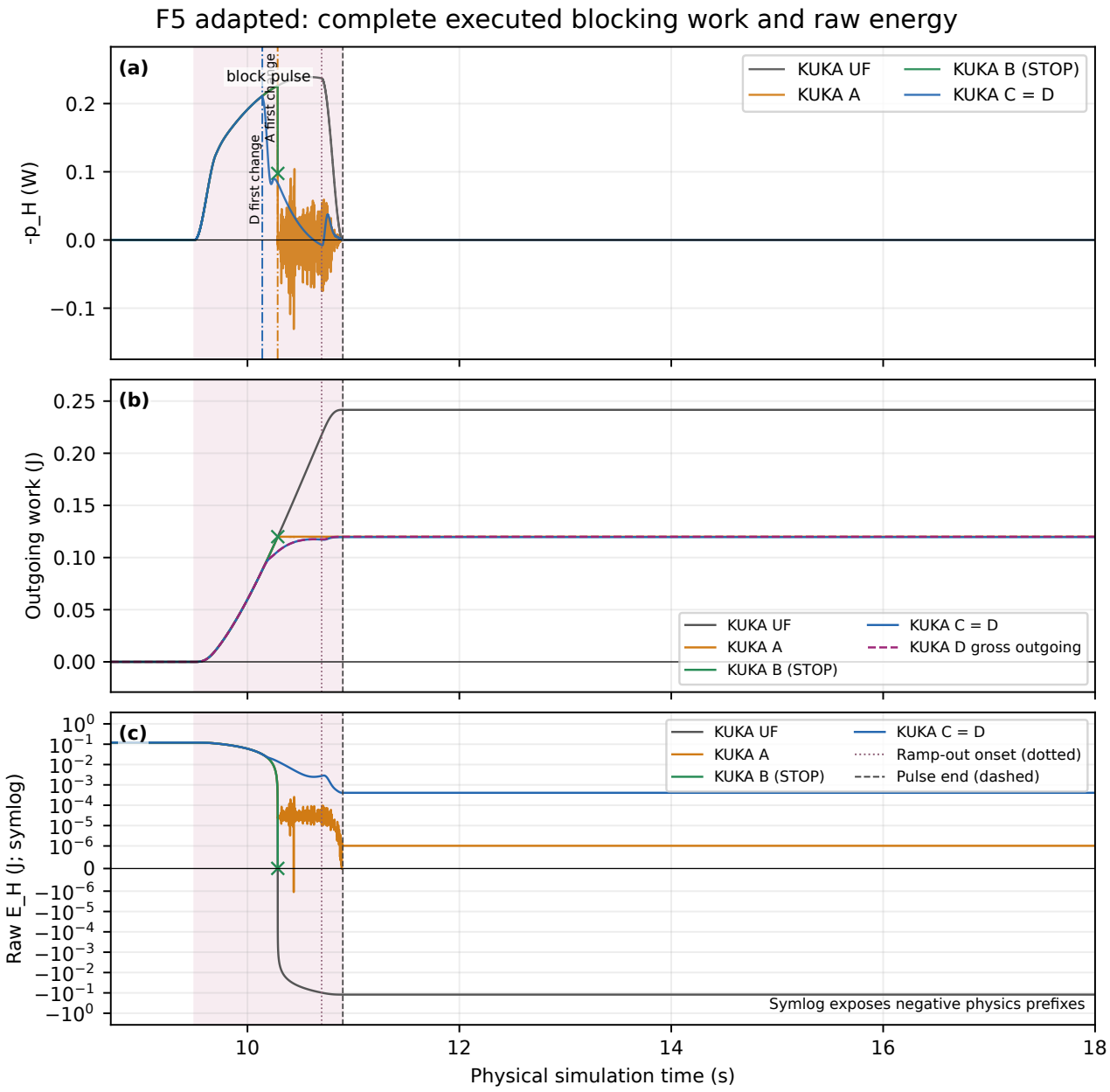
F4 and F6 — The current residual account is a copy



F4/F6. Supported accounting-identity diagnostic. Blocking traces, together with no-human F3, expose E_H = E_R because both executed accounts receive the same human work. The predictor shares the immutable plant model and holds the currently measured force at its known application point; mismatch power p_delta is not implemented or replaced with a zero-valued measurement.

The coincident accounts do not demonstrate mismatch accounting or the original p_R = p_H + p_delta decomposition. Reproducing F4/F6 requires future controller/model work on a physically consistent residual-work meter and independently configured plant/predictor parameters.

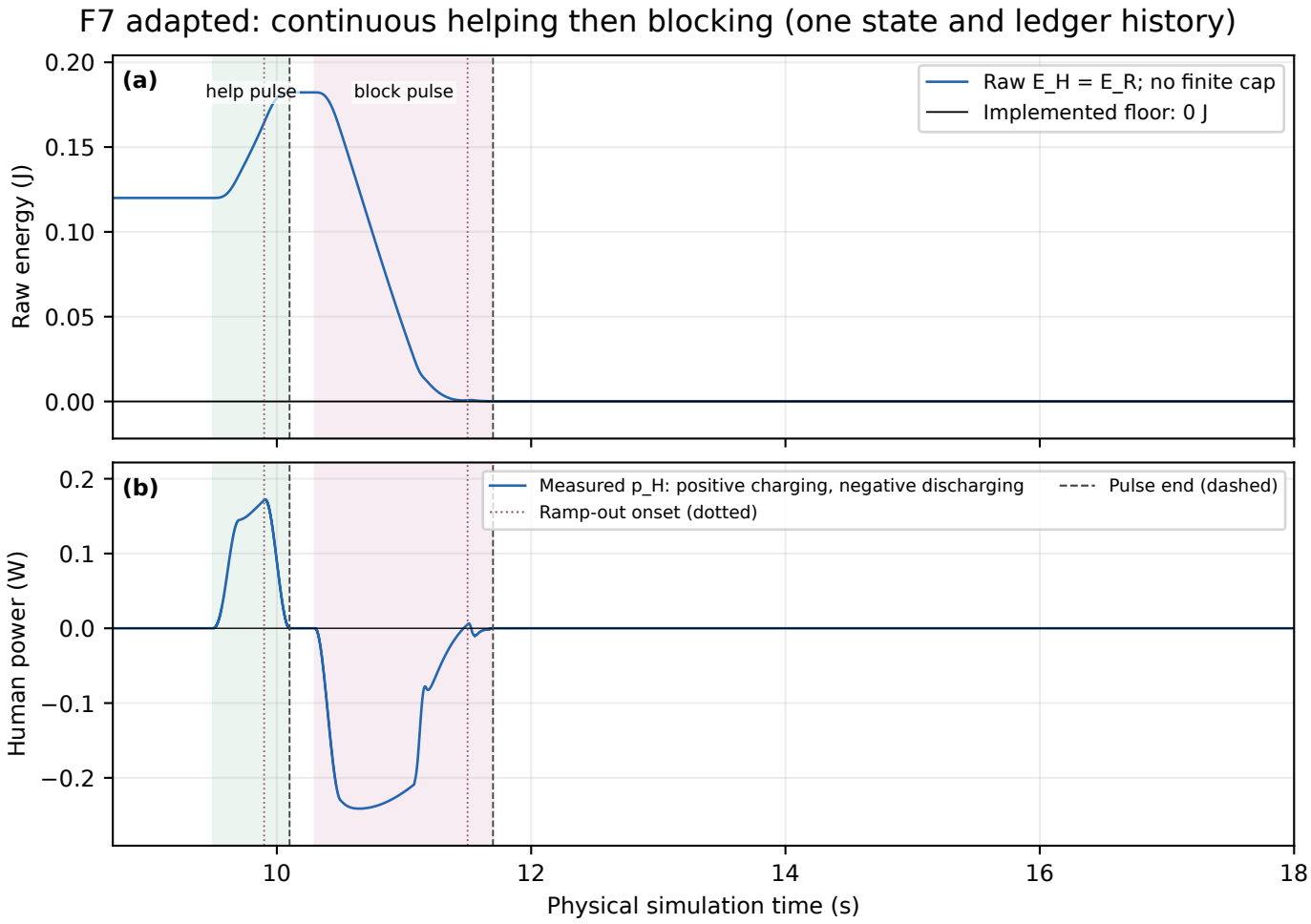
F5 — Blocking limits signed output, with failures retained



F5. Complete executed blocking power, work and raw energy. The panels show positive human-directed output -p_H, signed NET work with gross output identified separately, and raw energy against the implemented zero floor. Ramp-out onset, pulse end, actual intervention and premature termination are distinct events; no unenforced peak-power threshold is drawn.

Main D finishes with net output 0.119591 J, gross output 0.120004 J and minimum raw energy 0.0004092 J. A reaches a negative physics minimum of -1.11e-06 J despite endpoint checks, while B stops at 10.286 s; these are signed-energy results rather than peak-power, force or gross-work guarantees.

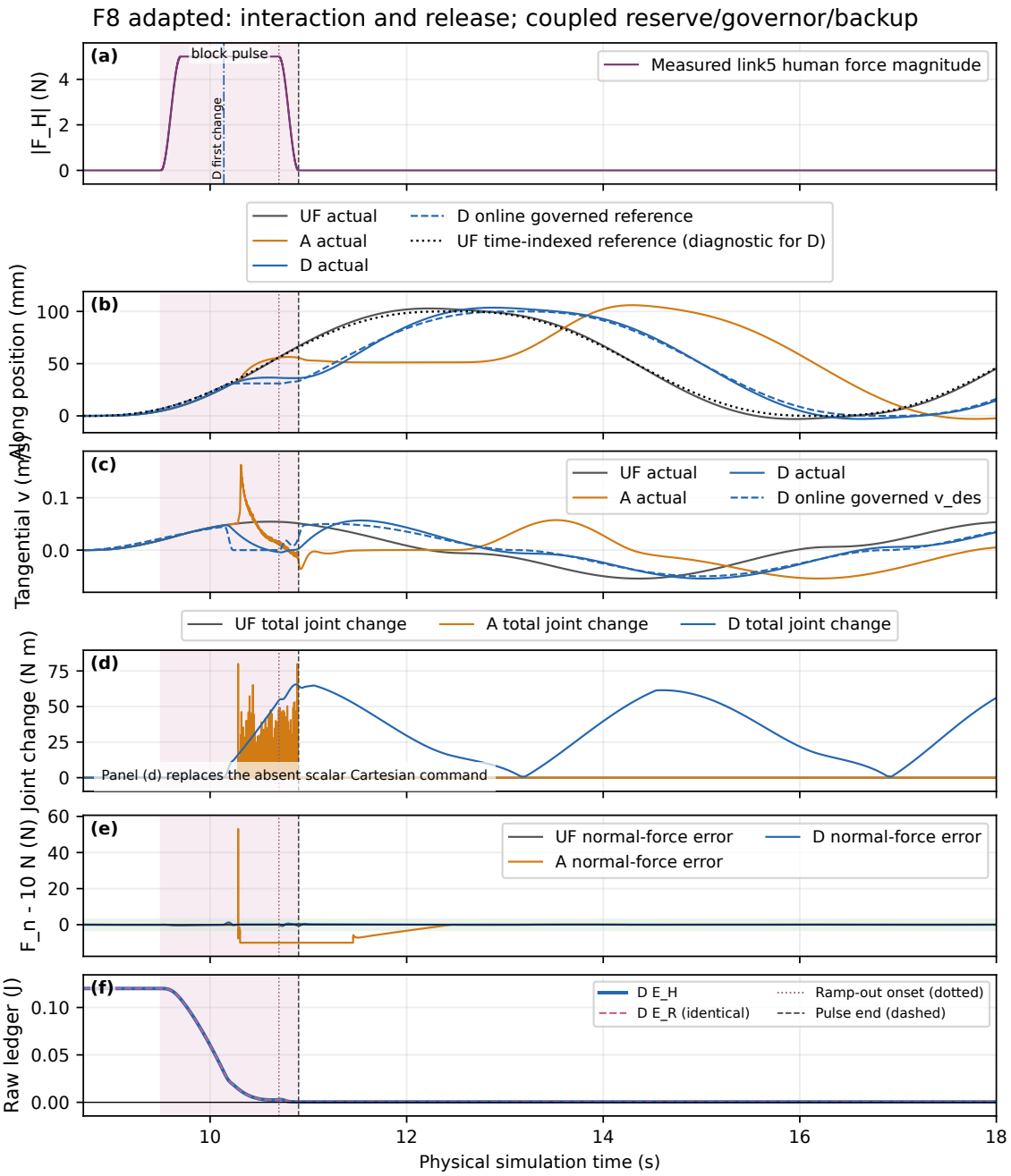
F7 — Charging and discharging in one continuous run



F7. Continuous help-then-block accounting. The frozen help-then-block profile runs continuously with the same plant, controller, reference history and uncapped ledgers, retaining the implemented zero floor.

Measured helping work is 0.062228 J, raising energy from 0.12 J to 0.182228 J without clipping; the repeated run matches all executed signals and initial/final states exactly. The run completes at 18 s with net outgoing 0.119849 J, gross outgoing 0.182297 J and minimum raw energy 0.000150544 J. Gross output can exceed the initial budget because measured positive human-point work recharges it, while the implemented constraint concerns signed net work; phase interpretation uses measured power rather than force direction alone.

F8 — Complete interaction and recovery of the current policy



F8. Adapted interaction/recovery comparison. Six panels retain force/events, actual and online governed position/velocity, original-target force error and raw H/R accounts, with any original-time reference marked diagnostic; panel (d) replaces the absent scalar Cartesian command with joint-torque intervention in N m.

D starts intervening at 10.14 s; A first intervenes at 10.285 s. Main D sustained force recovery takes 0.262 s from ramp-out onset, or 0.062 s from full force removal, using a full 0.5 s all-good window and explicitly distinguishing the old paper's ramp-out release anchor.

KUKA's virtual clock preserves stroke endpoints rather than capturing the old spatial anchor, and this coupled reserve/governor/backup comparison does not isolate a governor effect.

Quantitative task, interaction and recovery outcomes

Main policy	Force RMSE [N]	MAE [N]	Max error [N]	Absent [s]	End [s]
UF	0.074	0.04	0.397	0	18
A	3.838	1.691	53.104	1.14975	18
B	0.482	0.103	18.439	0	10.286
C	0.129	0.063	1.241	0	18
D	0.129	0.063	1.241	0	18

These force statistics use each run's executed task interval [8.71 s, actual end). Equal-duration main UF/A/D rows support full-task comparisons; B ends early and its full-run aggregate has a shorter window. No samples are repeated after termination. Maximum errors and contact absence retain the original 10 N target during controller recovery.

Common window	Force RMSE [N]	Force MAE [N]	Tangent RMSE [m/s]
UF	0.133	0.091	0.00257
A	1.341	0.125	0.00257
B	0.482	0.103	0.00257
C	0.231	0.129	0.00918
D	0.231	0.129	0.00918

The equal-support table ends at 10.286 s for every main policy. It does not replace the full-run completion and violation outcomes.

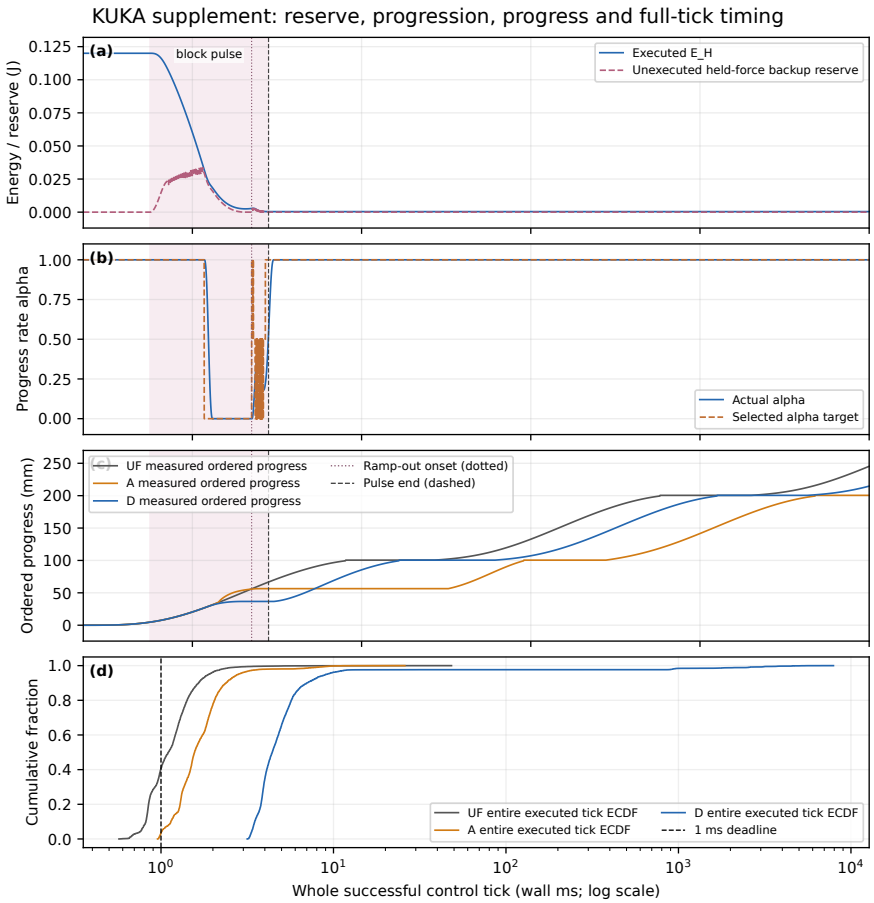
Run	Tangent RMSE [m/s]	Peak speed [m/s]	Online debt [mm]	Force recovery [s]
Main A	0.00893	0.163	5.976	1.49
Main D	0.00756	0.0567	5.62	0.062
Historical D	0.00826	0.0552	7.45	0.078

Run	Peak original-reference debt [mm]	Progress / matched UF [%]
Main A	77.982	81.73
Main D	32.756	87.526
Historical D	26.974	89.363

Run	Net / gross [J]	Peak output [W]	Progress [m]	Outcome
Main A	0.119999 / 0.124648	0.226	0.2005	complete
Main D	0.119591 / 0.120004	0.211	0.2147	complete
Historical A	0.119977 / 0.119977	0.217	0.0588	stopped
Historical D	0.119869 / 0.119927	0.217	0.2192	complete
Tiny budget D	6.86e-05 / 6.86e-05	0.006	0.0054	stopped

Tangential tracking compares actual TCP velocity with the online governed reference at control samples; the peak-speed metric uses physics samples. Online tracking debt, original-reference debt and ordered task-progress loss are distinct declared metrics. Force recovery is measured from pulse end and requires 0.5 s of sustained recovery. The historical A stop and tiny-budget D stop remain failures to complete, even when their executed prefixes satisfy a subset of constraints.

KUKA supplement — Reserve, progression, completion and runtime



KUKA-specific supplement. Available actual energy and the unexecuted current-held-force backup reserve have different provenance. Alpha target and actual alpha expose transition dynamics. Ordered progress avoids treating back-and-forth displacement as completed work. Whole successful control-tick timing is compared with the 1 ms simulated deadline; timing distributions are not a real-time certificate.

Main policy	Recorded QP calls	Torque correction ticks	Intervention ticks
UF	0	0	0
A	334	334	334
B	2	1	1
C	0	626	705
D	0	626	705

Zero recorded C/D solves or active rows on selected executed paths does not prove that no solver work occurred inside rejected plans: the existing guard discards some attempted-plan timing/diagnostics after a reserve or reaction-path rejection. No active task-QP advantage is inferred from this record.

Main D pauses for 0.482 s and accumulates maximum virtual-time delay 0.669 s. Its ordered measured progress at 18 s is 0.2147 m versus matched UF 0.2453 m. Preserved continuous extensions measure equal-progress completion without resetting the clock, ledgers or controller history.

Additional physical time to reach the matched UF 18 s task amount is main 0.668 s and historical 0.546 s; the reference amount and crossing rule are recorded in the manifest.

Main D successful whole-tick wall time: median 0.004415 s, p99 2.544035 s, maximum 7.916327 s; 18000 measured successful ticks exceed 1 ms. Full forecasts are offline computations, and failures are reported separately. Current measured force is held throughout each prediction; future scripted pulse values are not supplied to the guard.

Four-round verification, provenance and remaining limits

Round	Verdict	Recorded evidence
1	PASS	Canonical mapping and independent raw metrics passed; elapsed-zero timestamps, curve provenance and PDF legibility corrected. F7 was pending at this round and completed in Round 3.
2	PASS	All 13 matched fixtures and five transparent fixture families verified; 26 focused tests passed. F3 startup accounting and release-anchor labels corrected.
3	PASS	Reused exact-source main repeat and fresh no-human, stop and repeated continuous runs verified; 597 headline and 30 common-window checks passed. Charging coverage and torque mapping corrected.
4	PASS	Clean build and all 10 PDF pages inspected; four legend/annotation occlusions corrected. Metrics and curve contracts unchanged; scientific claims and preservation verified.

Detailed findings, justified corrections, commands, true subprocess exit status and structured test counts are in reviews/round_01.md through round_04.md and review_summary.json. Report correctness is assessed separately from scientific success: an unsupported mismatch mechanism, an inactive task-QP comparison and an explicit stop can be reported correctly.

Proposal component	Coverage
Known-point signed work; task exemption; finite fixture reserve/recovery	Demonstrated on reported fixtures
Continuous bidirectional charging without an upper cap	Demonstrated on the frozen continuous fixture and exact repeat
Per-stage task QP	Partially tested; no selected accepted task-QP rows in archived D matrix
Whole-port causal baseline; independent mismatch/residual account	Missing
Isolated governor ablation; finite ledger cap; peak-power/force protection	Missing
Global passivity; arbitrary-force robustness; recursive feasibility; 1 kHz hardware	Not established

All plotted physics powers and work increments remain unsmoothed. Actual raw ledgers are never lower-clamped; rejected candidates and unexecuted backup predictions are separated from executed trajectories. The complete manifest maps every curve to its preserved raw file/hash, source/configuration, units, sampling stage, transform and window. Original source, results and reports remain preserved in their prior worktrees.

Starting synchronized Stage 2 source: 41359f20430c930eec5afaebffa2cfbc482f07da. Numerical-source archive: 81dffa64109752a9a027975e7b831daa4ff91e86. New plotting branch: feature/kuka-poc-figures-20261016-41359f2. Final artifact/source hashes and the final Git revision are recorded in the reproducibility and preservation records, avoiding a self-referential commit hash in the PDF.

Reproduction: use the exact single-entry build command in reproduction.md with preserved inputs, a clean new output directory and the four review records. Simulation generation is separate and uses the frozen run configuration. The editable Markdown also preserves the complete command.